

KAIST
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[MAS456] Statistical Methods with Computer
Homework 1

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Contents

1	Chapter 2 - Statistical Learning	2
2	Chapter 3 - Linear Regression	9

Chapter 2 - Statistical Learning

(1) #2

- (a) **Data:** Information about the top 500 firms in the US, including profit, number of employees, industry, and CEO salary.

Objective: To understand which factors affect CEO salary.

Analysis:

Type: This is an **inference** problem because we want to understand the relationship between CEO salary and the other factors.

Classification or Regression: This is a **regression** problem because the outcome variable (CEO salary) is continuous.

n (number of observations): The number of observations would depend on how many firms are in the top 500 list. So, **n = 500**.

p (number of predictors): The number of predictors (independent variables) is **3** (profit, number of employees, industry).

- (b) **Data:** Information about 20 similar products, including whether they were a success or failure, price, marketing budget, competition price, and ten other variables.

Objective: To predict whether a new product will be a success or failure.

Analysis:

Type: This is a **prediction** problem because we want to predict the success or failure of a new product.

Classification or Regression: This is a **classification** problem because the outcome variable (success or failure) is categorical.

n (number of observations): **n = 20** because we have data on 20 similar products.

p (number of predictors): There are **13** predictors (price, marketing budget, competition price, and ten other variables).

- (c) **Data:** Weekly data for all of 2012, including the % change in the USD/Euro exchange rate, % changes in the US, British, and German markets.

Objective: To predict the % change in the USD/Euro exchange rate in relation to the weekly changes in the world stock markets.

Analysis:

Type: This is a **prediction** problem because we want to predict the % change in the USD/Euro exchange rate.

Classification or Regression: This is a **regression** problem because the outcome variable (% change) is continuous.

n (number of observations): The number of observations would depend on how many weeks are in 2012. Let's assume $n = 52$ (for 52 weeks).

p (number of predictors): There are **3** predictors (% change in the US market, % change in the British market, % change in the German market).

In summary:

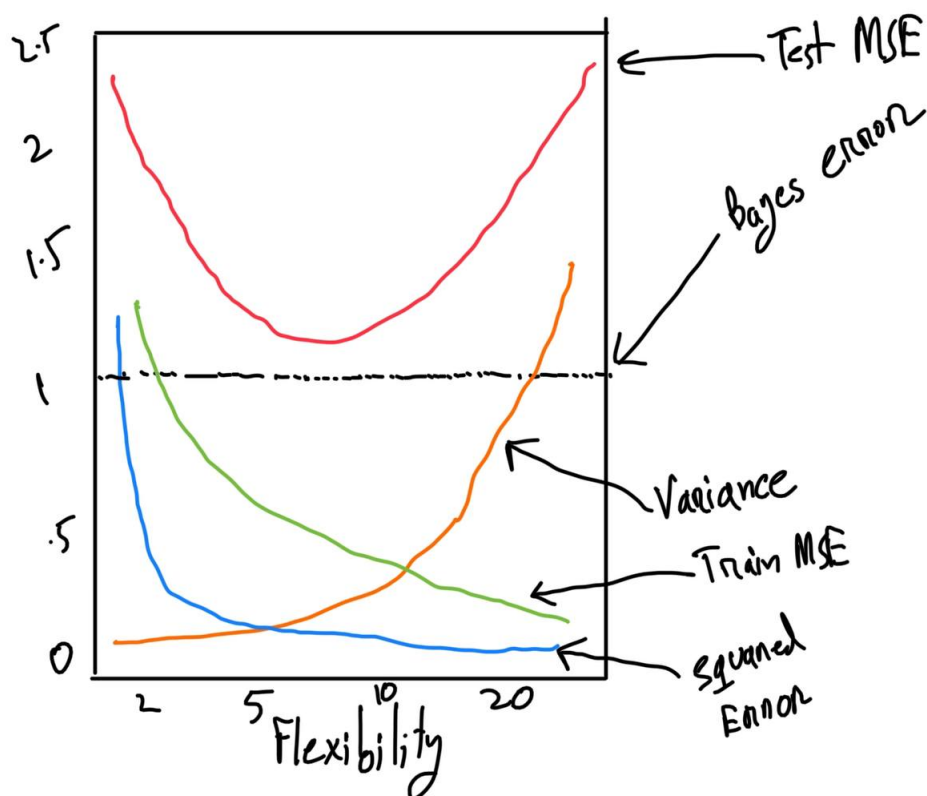
(a) **Regression, Inference, $n = 500$, $p = 3$**

(b) **Classification, Prediction, $n = 20$, $p = 13$**

(c) **Regression, Prediction, $n = 52$, $p = 3$**

(2) #3

- (a) In this sketch, we'll represent typical squared bias, variance, training error, test error, and Bayes (or irreducible) error curves as we move from less flexible to more flexible statistical learning methods on a single plot. The x-axis represents the amount of flexibility in the method, and the y-axis represents the values for each curve:



- (b) Explanation for the shape of each curve:

Squared Bias: Initially, with less flexible models, the bias is high because these models are too simple to capture the underlying patterns in the data. As we move towards more flexible models, bias decreases because they can fit the data better. However, if we go too far to the right on the x-axis (too much flexibility), the bias can start to increase again as

models overfit the noise in the data.

Variance: Initially, with less flexible models, the variance is low because they are less sensitive to fluctuations in the training data. As we move towards more flexible models, variance increases because they become more sensitive to small variations in the data. Very flexible models can exhibit extremely high variance as they fit noise in the data.

Training Error: The training error decreases monotonically as we move from less flexible to more flexible models. More flexible models can perfectly fit the training data, resulting in a training error of zero.

Test Error: The test error initially decreases as we move from less flexible to moderately flexible models because the models are better at capturing the underlying patterns in the data. However, beyond a certain point, the test error starts to increase due to overfitting. The test error is typically U-shaped, indicating an optimal level of model complexity.

Irreducible Error (Bayes Error): This is the inherent noise or variability in the data that cannot be reduced by any model. It remains constant and does not depend on the model's flexibility.

The key idea is that there is a trade-off between bias and variance. Less flexible models have high bias and low variance, while more flexible models have low bias and high variance. The goal is to find the right level of model complexity that minimizes the test error, which is the sum of bias and variance. This trade-off is often referred to as the bias-variance trade-off, and the test error curve represents it.

(3) #5

Advantages of a Very Flexible Approach:

Better Fit to Complex Data Patterns: Very flexible models, such as high-degree polynomial regression or deep neural networks, can capture highly complex and nonlinear relationships in the data. This can lead to more accurate predictions when the underlying data patterns are intricate.

High Training Accuracy: Flexible models can achieve very low training error, sometimes even zero. This means they can closely approximate the training data, potentially resulting in a model that can explain most of the variance in the data.

Disadvantages of a Very Flexible Approach:

Overfitting: The primary drawback of very flexible models is their susceptibility to overfitting. They can fit not only the underlying patterns but also the noise in the data. This leads to poor generalization to unseen data, resulting in high test error.

Complexity and Interpretability: Highly flexible models are often complex, making them difficult to interpret and understand. They may not provide insights into the underlying relationships in the data.

Computational Complexity: Training and evaluating very flexible models can be computationally expensive and time-consuming, especially for large datasets.

When a More Flexible Approach Might Be Preferred:

Complex Data Relationships: When there are complex, nonlinear relationships between predictors and outcomes, a more flexible approach can be preferred. It allows the model to capture these intricate patterns.

High-Dimensional Data: In high-dimensional datasets with many features, flexible models can help uncover hidden patterns that simpler models might miss.

Data Augmentation: When data augmentation techniques (such as feature engineering or deep learning) can help create informative features from raw data, more flexible models can better leverage these features.

When a Less Flexible Approach Might Be Preferred:

Limited Data: In situations with limited data, it's often wise to use simpler, less flexible models to avoid overfitting. These models have higher bias but lower variance and are less likely to fit noise.

Interpretability: When interpretability and model transparency are crucial, as in medical or legal contexts, less flexible models like linear regression or decision trees are preferred because they are easier to explain.

Computation Constraints: In applications with limited computational resources, such as real-time systems or edge devices, less flexible models are preferred due to their lower computational complexity.

Stability: Simpler models are less sensitive to changes in the data and are often more stable over time. In applications where model stability is critical, less flexible models may be preferred.

In practice, the choice between a very flexible and a less flexible approach depends on the specific problem, the available data, computational resources, and the balance between model accuracy and interpretability. Often, a middle-ground approach that balances flexibility and simplicity, such as ensemble methods or regularized models, can offer a good compromise.

(4) #8

(h) I have explored some of the statistics of the quantitative variables in the dataset:

Table 1: Summary Statistics for Quantitative Variables

Variable	Mean	Median	Mode
Apps	3001.638	1558.0	440.0
Enroll	779.973	434.0	177.0
F.Undergrad	3699.907	1707.0	500.0
Expend	9660.171	8377.0	4900.0

(5) #9

(a) **Quantitative Predictors:** ['mpg', 'cylinders', 'displacement', 'weight', 'acceleration', 'year', 'origin']

Qualitative Predictors: ['horsepower', 'name']

(b) The range of quantitative predictors is as follows:

Predictor	Minimum	Maximum
mpg	9.0	46.6
cylinders	3	8
displacement	68.0	455.0
weight	1613	5140
acceleration	8.0	24.8
year	70	82
origin	1	3

(c) The mean and standard deviation of quantitative predictors are as follows:

Predictor	Mean	Standard Deviation
mpg	23.515869	7.825804
cylinders	5.458438	1.701577
displacement	193.532746	104.379583
weight	2970.261965	847.904119
acceleration	15.555668	2.749995
year	75.994962	3.690005
origin	1.574307	0.802549

- (d) The range, mean, and standard deviation of quantitative predictors in the subset of the data are as follows:

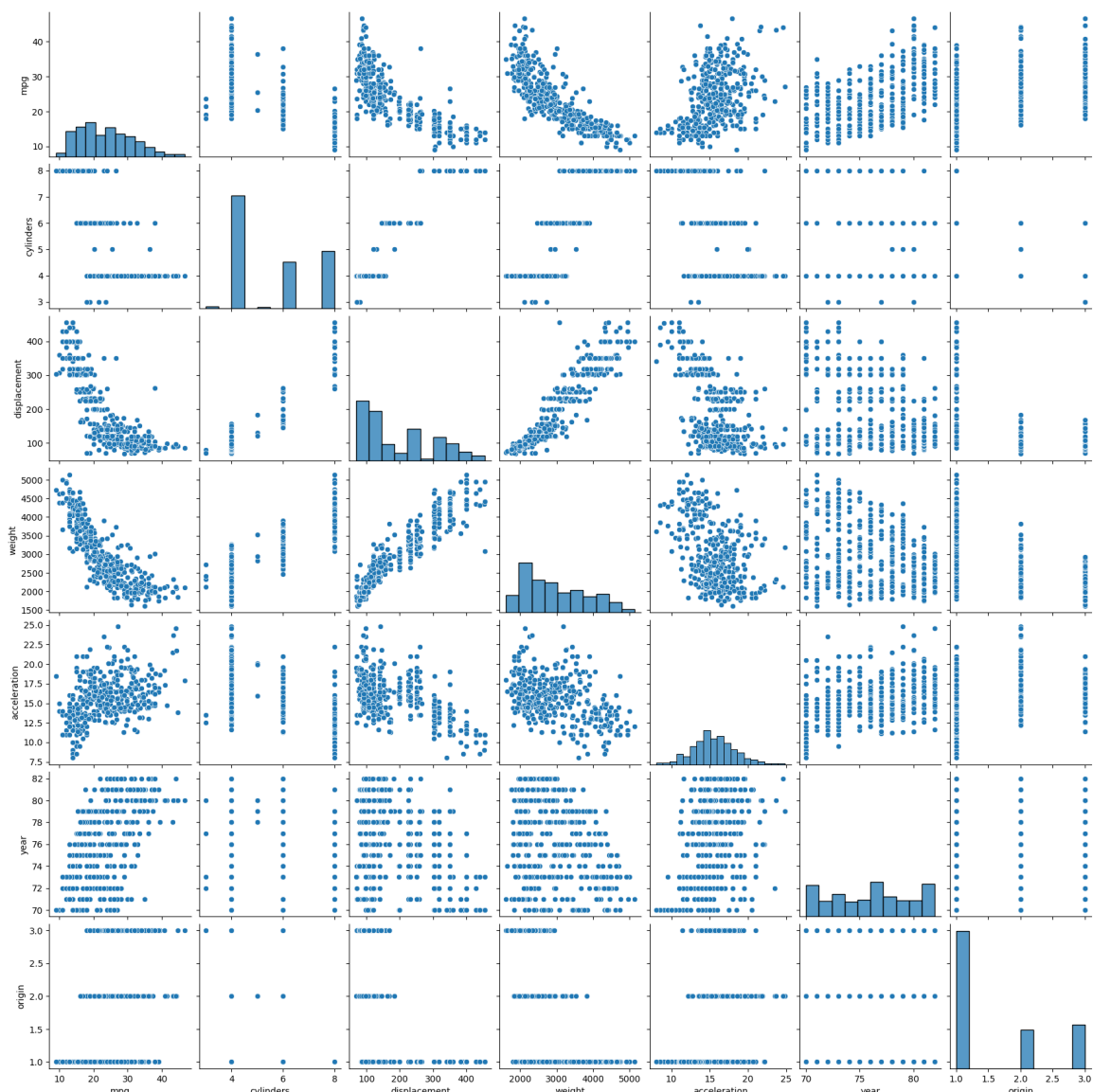
Range of Quantitative Predictors (Subset):

Predictor	Minimum	Maximum
mpg	11.0	46.6
cylinders	3	8
displacement	68.0	455.0
weight	1649	4997
acceleration	8.5	24.8
year	70	82
origin	1	3

Mean and Standard Deviation of Quantitative Predictors (Subset):

Predictor	Mean	Standard Deviation
mpg	24.438629	7.908184
cylinders	5.370717	1.653486
displacement	187.049844	99.635385
weight	2933.962617	810.642938
acceleration	15.723053	2.680514
year	77.152648	3.111230
origin	1.598131	0.816163

- (e) We can see different types of correlations among the variables. some are highly positively correlated while others are highly negatively correlated. For example, displacement and weight seem positively correlated while mpg and displacement are negatively correlated.



- (f) The correlations between the predictors and the target variable (mpg) are as follows:

Predictor	Correlation with MPG
mpg	1.000000
cylinders	-0.776260
displacement	-0.804443
weight	-0.831739
acceleration	0.422297
year	0.581469
origin	0.563698

So we can use variables that have high absolute values for correlation such as weight and displacement to predict mpg.

Chapter 3 - Linear Regression

(6) #1

In Table 3.4, we have the p-values associated with the coefficients of a multiple linear regression model, where we are trying to predict the number of units sold based on advertising budgets for TV, radio, and newspaper. Let's describe the null hypotheses corresponding to these p-values and explain the conclusions we can draw:

Null Hypothesis for Intercept (Intercept coefficient): The null hypothesis is that when all advertising budgets (TV, radio, and newspaper) are zero, the expected number of units sold is equal to zero. In other words, the null hypothesis states that advertising budgets have no effect on sales.

Conclusion: The p-value for the Intercept is less than 0.0001 (very low). Therefore, we can reject the null hypothesis for the Intercept. This suggests that there is a statistically significant relationship between some combination of advertising budgets (TV, radio, and newspaper) and sales.

Null Hypothesis for TV Advertising (TV coefficient): The null hypothesis is that there is no significant linear relationship between TV advertising spending and sales. Mathematically, it implies that the coefficient for TV advertising in the linear regression model is equal to zero.

Conclusion: The p-value for TV advertising is less than 0.0001 (very low). We can reject the null hypothesis for TV advertising. This indicates that TV advertising spending has a statistically significant effect on sales. In other words, there is strong evidence to suggest that TV advertising affects the number of units sold.

Null Hypothesis for Radio Advertising (Radio coefficient): The null hypothesis is that there is no significant linear relationship between radio advertising spending and sales. Mathematically, it implies that the coefficient for radio advertising in the linear regression model is equal to zero.

Conclusion: The p-value for radio advertising is less than 0.0001 (very low). We can reject the null hypothesis for radio advertising. This indicates that radio advertising spending has a statistically significant effect on sales. In other words, there is strong evidence to suggest that radio advertising affects the number of units sold.

Null Hypothesis for Newspaper Advertising (Newspaper coefficient): The null hypothesis is that there is no significant linear relationship between newspaper advertising spending and sales. Mathematically, it implies that the coefficient for newspaper advertising in the linear regression model is equal to zero.

Conclusion: The p-value for newspaper advertising is 0.8599 (high). We fail to reject the null hypothesis for newspaper advertising. This suggests that newspaper advertising spending does not have a statistically significant effect on sales based on the data and model used. In other words, there is insufficient evidence to conclude that newspaper advertising affects the number of units sold.

In summary, based on the p-values:

TV advertising and radio advertising budgets have a statistically significant effect on sales, and there is strong evidence to support this conclusion.

Newspaper advertising budget, on the other hand, does not have a statistically significant effect on sales, and there is insufficient evidence to support a relationship between newspaper advertising and the number of units sold.

(7) #4

- (a) We fit both a linear regression model and a cubic regression model, i.e., $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$.

For the training residual sum of squares (RSS):

- **Linear Regression (RSS_{linear}):** The linear model should be able to capture the linear relationship between X and Y effectively. However, because of noises present in the data, we expect the training RSS for the linear regression (RSS_{linear}) to be higher compared to the cubic regression model because of good fitting.

- **Cubic Regression (RSS_{cubic}):** The cubic regression model is more flexible and can fit more complex relationships. However, in this case, the true relationship is linear. The cubic regression model may fit the training data very closely and is likely to overfit, leading to a lower training RSS (RSS_{cubic}) compared to the linear regression.

- (b) For the test RSS (rather than training RSS), the expectation becomes the opposite as compared to part (a):

- We expect the test RSS for the linear regression (RSS_{linear}) to be lower compared to the cubic regression model (RSS_{cubic}) because the true relationship is linear. The linear model is more likely to generalize well to unseen data.

(c) For the training RSS:

- **Linear Regression (RSS_linear):** The linear model assumes a linear relationship. If the true relationship is not far from linear, the linear model may perform reasonably well and result in a low training RSS (RSS_{linear}), but it can not outperform cubic regression.
- **Cubic Regression (RSS_cubic):** The cubic regression model is more flexible and can fit nonlinear relationships as well as linear relationships. So the cubic regression will outperform the linear regression in any case, resulting in a lower training RSS (RSS_{cubic}).

(d) For the test RSS:

- **Linear Regression (RSS_linear):** The linear model may generalize better to the test data if the true relationship is not far from linear. Therefore, we might expect a lower test RSS (RSS_{linear}).
- **Cubic Regression (RSS_cubic):** The cubic regression model, being more flexible, could perform well on the test data if the true relationship has nonlinear aspects. Thus, it's possible that the cubic regression has a lower test RSS (RSS_{cubic}).

There is not enough information to definitively conclude which one will be lower in terms of test RSS because it depends on the specific nature of the true relationship and the test data.

(8) #7

We aim to demonstrate the equality between R^2 and r^2 in the context of simple linear regression. We can do a general proof using Cov and Var:

$$\begin{aligned}
 R^2 &= \frac{TSS - RSS}{TSS} \quad (\text{by definition}) \\
 &= \frac{ESS}{TSS} \quad (\text{by definition}) \\
 &= \frac{\text{Var}(\hat{y})}{\text{Var}(y)} \quad (\text{expressing ESS and TSS in terms of variance}) \\
 &= \frac{\text{Var}(b_0 + b_1 X)}{\text{Var}(y)} \quad (\text{substituting the linear regression model}) \\
 &= b_1^2 \cdot \frac{\text{Var}(X)}{\text{Var}(y)} \quad (\text{factoring out } b_1^2) \\
 &= \left(\frac{\text{Cov}(X, y)}{\text{Var}(X)} \right)^2 \frac{\text{Var}(X)}{\text{Var}(y)} \quad (\text{using covariance}) \\
 &= \frac{\text{Cov}(X, y)^2}{\text{Var}(X)\text{Var}(y)} \quad (\text{simplifying}) \\
 &= r^2 \quad (\text{since } r \text{ is the correlation coefficient})
 \end{aligned}$$

Hence, we've established that R^2 is indeed equivalent to r^2 within the framework of simple linear regression.

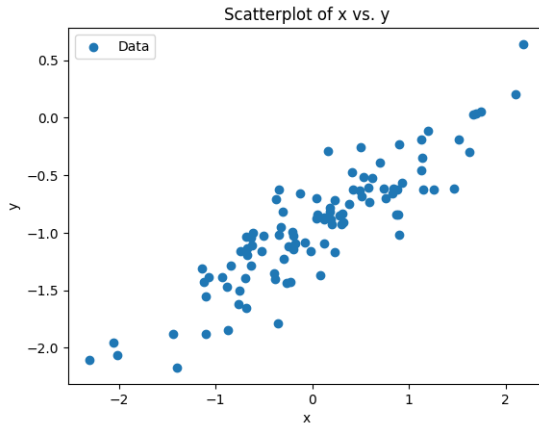
(9) #13

(a) Done

(b) Done

(c) The length of vector y is 100. The values of β_0 and β_1 in this linear model are -1 and 0.5, respectively.

(d) We created a scatterplot displaying the relationship between x and y . We can observe a linear relationship between the two variables.



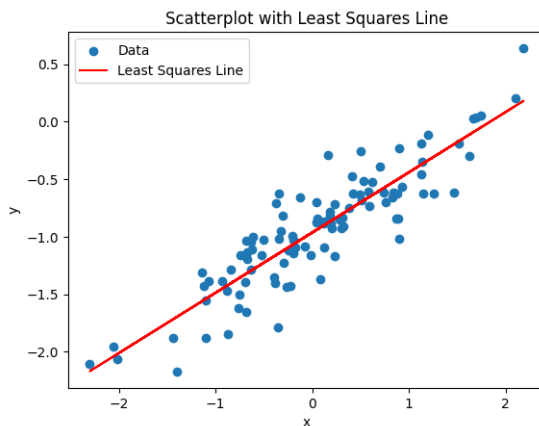
(e) We fitted a least squares linear model to predict y using x . The estimated coefficients are as follows:

$$\hat{\beta}_0 \text{ (intercept)} : -0.9632$$

$$\hat{\beta}_1 \text{ (slope)} : 0.5239$$

These coefficients represent the best-fit line that minimizes the sum of squared residuals. They approximate the true coefficients β_0 and β_1 in the linear model.

(f) Here it is:



(g) We computed the R-squared values to assess the goodness of fit for the models:

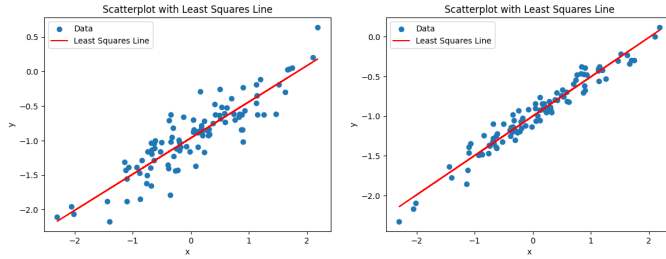
Linear Model The R-squared value for the least squares linear model is $R^2 = 0.7997$. This indicates that approximately 79.97% of the variance in the dependent variable y is explained by the linear relationship with the independent variable x .

Polynomial Model Similarly, for the polynomial regression model that includes both x and x^2 , the R-squared value is $R^2 = 0.7998$. This demonstrates a similar level of explanatory power as the linear model.

These R-squared values suggest that both the models provide a good fit to the data, explaining a significant portion of the variance in y . There is no significant improvement in using a polynomial model.

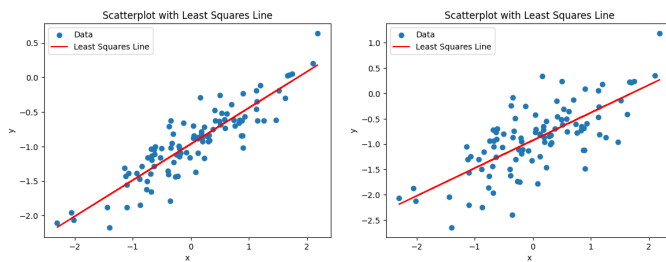
- (h) These are the plots for variance 0.25 and 0.1.

We reduced the variance of the error term ϵ , resulting in less noise in the data, and we observed that the R-squared value increased compared to the original data. This increase in R-squared indicates that the linear and polynomial models became even more effective at explaining the variance in the data when the noise level decreased. The higher R-squared values in this case suggest that the models provided an excellent fit to the data with reduced noise.



- (i) These are the plots for variance 0.25 and 0.5.

We increased the variance of ϵ to introduce more noise into the data, the R-squared values decreased significantly compared to the original data. This substantial decrease in R-squared values signifies the challenges that the models faced in explaining the variance in the presence of increased noise. The lower R-squared values in this scenario indicate that the models struggled to capture the underlying relationships between the variables due to the presence of considerably more noise.



- (j) **Original Data:**

$$\begin{bmatrix} 95\% \text{ Confidence Interval for } \beta_0 : [-1.0039, -0.9666] \\ 95\% \text{ Confidence Interval for } \beta_1 : [0.4885, 0.5306] \end{bmatrix}$$

Comments: In the original data set, the 95% confidence interval for β_0 ranges from approximately -1.0039 to -0.9666, while the interval for β_1 ranges from about 0.4885 to 0.5306. These intervals provide a range of values where we are reasonably confident that the true population coefficients β_0 and β_1 lie. The narrow width of these intervals suggests relatively precise estimates of the coefficients.

Noisier Data:

$$\begin{bmatrix} 95\% \text{ Confidence Interval for } \beta_0 : & [-1.0947, -0.8930] \\ 95\% \text{ Confidence Interval for } \beta_1 : & [0.3658, 0.5932] \end{bmatrix}$$

Comments: In the data with increased noise, the 95% confidence interval for β_0 widened significantly, spanning from approximately -1.0947 to -0.8930. Similarly, the interval for β_1 also became wider, ranging from about 0.3658 to 0.5932. This indicates that in the presence of more noise, our confidence in the coefficient estimates decreases, resulting in less precise parameter estimates.

Less Noisy Data:

$$\begin{bmatrix} 95\% \text{ Confidence Interval for } \beta_0 : & [-1.0227, -0.9801] \\ 95\% \text{ Confidence Interval for } \beta_1 : & [0.4654, 0.5134] \end{bmatrix}$$

Comments: When we reduced the noise in the data, the 95% confidence interval for β_0 remained similar in width, ranging from approximately -1.0227 to -0.9801. However, the interval for β_1 also remained relatively narrow, ranging from about 0.4654 to 0.5134. This suggests that with less noisy data, our confidence in the coefficient estimates remains high, resulting in precise parameter estimates.

(10) #14

- (a) The form of the linear model is: $y = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \varepsilon$

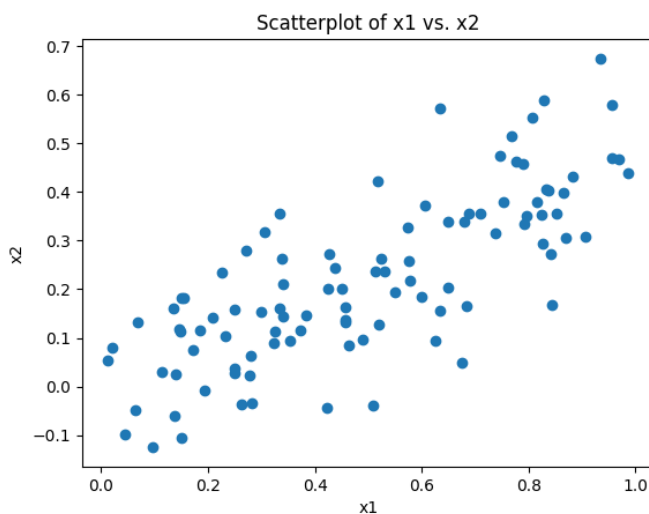
The regression coefficients are:

$$\beta_0 = 2$$

$$\beta_1 = 2$$

$$\beta_2 = 0.3$$

- (b) The correlation between x_1 and x_2 is approximately 0.772.



- (c) After fitting a least squares regression using only x_1 , the estimated coefficients are:

$$\hat{\beta}_0 \approx 1.958$$

$$\hat{\beta}_1 \approx 1.615$$

$$\hat{\beta}_2 \approx 0.943$$

The p-value for the null hypothesis $H_0 : \beta_1 = 0$ is approximately 0.003, which suggests that we can reject the null hypothesis and conclude that β_1 is not equal to zero.

The p-value for the null hypothesis $H_0 : \beta_2 = 0$ is approximately 0.259, which suggests that we fail to reject the null hypothesis, indicating that β_2 may be close to zero.

- (d) After fitting a least squares regression using only x_1 , the estimated coefficients are:

$$\hat{\beta}_0 \approx 1.937$$

$$\hat{\beta}_1 \approx 2.077$$

The p-value for the null hypothesis $H_0 : \beta_1 = 0$ is very close to zero (1.37×10^{-8}), indicating strong evidence to reject the null hypothesis, and suggesting that β_1 is significantly different from zero.

- (e) After fitting a least squares regression using only x_2 , the estimated coefficients are:

$$\hat{\beta}_0 \approx 2.324$$

$$\hat{\beta}_1 \approx 2.910$$

The p-value for the null hypothesis $H_0 : \beta_1 = 0$ is very close to zero (7.43×10^{-7}), indicating strong evidence to reject the null hypothesis, and suggesting that β_1 is significantly different from zero.

- (f) The results obtained in parts (c), (d), and (e) do not necessarily contradict each other. The significance of the coefficients and the ability to reject the null hypotheses depend on whether both x_1 and x_2 are included together in the model or not.

In part (c), when both x_1 and x_2 were included in the model simultaneously, we found that β_1 was a significant predictor of y .

In part (d), when only x_1 was included in the model, β_1 was still significant, and β_2 was not.

In part (e), when only x_2 was included in the model, β_1 was significant, and β_2 was not.

These results indicate that the significance of the coefficients depends on the presence of other variables in the model. The inclusion of different predictor variables can lead to dif-

ferent conclusions about the significance of individual coefficients.

- (g) After obtaining one additional observation and adding it to each of x_1 , x_2 , and y , we refitted the linear models from parts (c), (d), and (e) using this new data.

For the model in part (c), which includes both x_1 and x_2 , the new observation had the following effects:

- Outlier status: 2.784 (indicating that the new observation is considered an outlier)
- High-leverage status: 0.653 (indicating that the new observation has high leverage)

For the model in part (d), which includes only x_1 , the new observation had the following effects:

- Outlier status: 0.006 (indicating that the new observation is not considered an outlier)
- High-leverage status: 0.348 (indicating that the new observation has some leverage)

For the model in part (e), which includes only x_2 , the new observation had the following effects:

- Outlier status: 0.652 (indicating that the new observation is not considered an outlier)
- High-leverage status: 0.348 (indicating that the new observation has some leverage)

In each model, the new observation is characterized as an outlier in the model that includes both x_1 and x_2 (part c), but it is not considered an outlier in the models that include only x_1 (part d) and only x_2 (part e). Additionally, the new observation is found to have high leverage in all three models.

These findings suggest that the new observation has a significant impact on the model that includes both predictor variables, making it an outlier in that context. However, its influence is less pronounced when considering individual predictor variables separately.